

CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 13: Factorisation

Topic-wise short notes with solved examples

Factorising algebraic expressions by taking out common factors, by grouping, by using identities and by splitting the middle term of a trinomial.

## 1 Factors

- To **factorise** an expression is to write it as a **product** of simpler expressions. Each one is a **factor**.
- Factorising is the reverse of multiplying:  $3x(x + 2) = 3x^2 + 6x$ , so  $3x^2 + 6x = 3x(x + 2)$ .
- Check any factorisation by multiplying the factors back.
- Always take out the common factor first**, then look at what is left.

## 2 Common factors

- Find the HCF of the terms: the HCF of the numerical coefficients and the lowest power of each common variable.
- Write the HCF outside a bracket and divide each term by it.

**Example 2.1** Factorise  $6x^2y - 9xy^2 + 3xy$ .

SOLUTION

$$\text{HCF} = 3xy$$

$$3xy(2x - 3y + 1)$$

**Example 2.2** Factorise  $5(a + b) - 2x(a + b)$ .

SOLUTION

The bracket  $(a + b)$  is common.

$$(a + b)(5 - 2x)$$

**Example 2.3** Factorise  $x(a - 3) + y(3 - a)$ .

SOLUTION

$$3 - a = -(a - 3), \text{ giving } x(a - 3) - y(a - 3).$$

$$(a - 3)(x - y)$$

### 3 Grouping

- Used for four (or more) terms with no factor common to all.
- Arrange the terms in groups so that each group has a common factor and the same bracket is left in each.

**Example 3.1** Factorise  $ab + bc + ax + cx$ .

SOLUTION

$$b(a + c) + x(a + c)$$

$$(a + c)(b + x)$$

**Example 3.2** Factorise  $x^2 - 3x - xy + 3y$ .

SOLUTION

$$x(x - 3) - y(x - 3)$$

$$(x - 3)(x - y)$$

**Example 3.3** Factorise  $6ab - 4b + 6 - 9a$ .

SOLUTION

$$2b(3a - 2) - 3(3a - 2)$$

$$(3a - 2)(2b - 3)$$

### 4 Difference of two squares

$$a^2 - b^2 = (a + b)(a - b)$$

- Each term must be a perfect square and the sign between them must be minus.
- After using the identity, check whether a factor can be factorised again.

**Example 4.1** Factorise  $49x^2 - 81y^2$ .

SOLUTION

$$(7x + 9y)(7x - 9y)$$

**Example 4.2** Factorise  $3a^3 - 48a$ .

SOLUTION

$$3a(a^2 - 16)$$

$$3a(a + 4)(a - 4)$$

**Example 4.3** Factorise  $x^4 - 1$ .

SOLUTION

$$(x^2 + 1)(x^2 - 1)$$

$$(x^2 + 1)(x + 1)(x - 1)$$

**Example 4.4** Factorise  $(a + b)^2 - c^2$ .

SOLUTION

$$(a + b + c)(a + b - c)$$

**Example 4.5** Evaluate  $(73)^2 - (27)^2$ .

SOLUTION

$$(73 + 27)(73 - 27) = 100 \times 46$$

$$4600$$

## 5 Perfect square trinomials

- $a^2 + 2ab + b^2 = (a + b)^2$  and  $a^2 - 2ab + b^2 = (a - b)^2$ .
- **Test:** the first and last terms are perfect squares and the middle term is twice the product of their square roots.

**Example 5.1** Factorise  $9x^2 + 24xy + 16y^2$ .

SOLUTION

$$(3x)^2 + 2(3x)(4y) + (4y)^2$$

$$(3x + 4y)^2$$

**Example 5.2** Factorise  $4a^2 - 20a + 25$ .

SOLUTION

$$(2a - 5)^2$$

**Example 5.3** Factorise  $x^2 + 6x + 9 - y^2$ .

SOLUTION

$$(x + 3)^2 - y^2$$

$$(x + 3 + y)(x + 3 - y)$$

## 6 Trinomials of the form $x^2 + bx + c$

- Find two numbers whose **product is**  $c$  and whose **sum is**  $b$ . If they are  $p$  and  $q$ , then  $x^2 + bx + c = (x + p)(x + q)$ .

| Sign of $c$ | Sign of $b$ | Signs of $p$ and $q$                   |
|-------------|-------------|----------------------------------------|
| +           | +           | Both positive                          |
| +           | −           | Both negative                          |
| −           | +           | Opposite signs; the larger is positive |
| −           | −           | Opposite signs; the larger is negative |

**Example 6.1** Factorise  $x^2 + 9x + 20$ .

**SOLUTION**

Product 20, sum 9: 4 and 5.

$$(x + 4)(x + 5)$$

**Example 6.2** Factorise  $x^2 - 7x + 12$ .

**SOLUTION**

Product 12, sum  $-7$ :  $-3$  and  $-4$ .

$$(x - 3)(x - 4)$$

**Example 6.3** Factorise  $x^2 + 2x - 15$ .

**SOLUTION**

Product  $-15$ , sum 2: 5 and  $-3$ .

$$(x + 5)(x - 3)$$

**Example 6.4** Factorise  $y^2 - 4y - 21$ .

**SOLUTION**

Product  $-21$ , sum  $-4$ :  $-7$  and 3.

$$(y - 7)(y + 3)$$

## 7 Trinomials of the form $ax^2 + bx + c$

- Find two numbers whose **product is**  $ac$  and whose **sum is**  $b$ . Split the middle term with them and factorise by grouping.

**Example 7.1** Factorise  $2x^2 + 7x + 3$ .

**SOLUTION**

Product 6, sum 7: 6 and 1.  $2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3)$ .

$$(x + 3)(2x + 1)$$

**Example 7.2** Factorise  $3x^2 - 10x + 8$ .

**SOLUTION**

Product 24, sum  $-10$ :  $-6$  and  $-4$ .  $3x^2 - 6x - 4x + 8 = 3x(x - 2) - 4(x - 2)$ .

$$(x - 2)(3x - 4)$$

**Example 7.3** Factorise  $6a^2 - a - 2$ .

**SOLUTION**

Product  $-12$ , sum  $-1$ :  $-4$  and  $3$ .  $6a^2 - 4a + 3a - 2 = 2a(3a - 2) + 1(3a - 2)$ .

$$(3a - 2)(2a + 1)$$

## 8 Using factors to divide

- To divide one expression by another, factorise both and cancel the common factors.

**Example 8.1** Divide  $x^2 - 5x + 6$  by  $x - 2$ .

**SOLUTION**

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

$$x - 3$$

**Example 8.2** Simplify  $\frac{4x^2 - 9}{2x^2 + 3x}$ .

**SOLUTION**

$$\frac{(2x + 3)(2x - 3)}{x(2x + 3)}$$

$$\frac{2x - 3}{x}$$

## 9 Common mistakes

- Leaving a common factor inside the bracket.
- Sign errors when the second group starts with a minus.
- Trying to factorise a sum of squares such as  $x^2 + 9$ .
- For  $ax^2 + bx + c$ , looking for numbers with product  $c$  instead of  $ac$ .
- Stopping at  $(x^2 + 1)(x^2 - 1)$  without factorising  $x^2 - 1$ .
- Not checking by multiplying the factors back.

MathSetu